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Simulating ice–wave interactions in the Laurentian Great Lakes using a fully coupled hydrodynamic–ice–wave model

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ABSTRACT

Hydrodynamic modeling in cold climate regions necessitates more sophisticated approaches that accurately simulate ice–wave interactions. Traditional models often overlook the complex coupling mechanisms between ice and ocean waves, especially the two-way processes where ice attenuates wave energy and waves break ice floes. This oversight can also intensify modeling challenges in coastal areas, including large lakes, where ice–wave interactions influence storm surges, high waves, and coastal erosion. To address this gap, this paper introduces an enhanced modeling approach that integrates both ice-induced wave attenuation and wave-induced ice breakage. To implement these processes, the Finite-Volume Community Ocean Model (FVCOM) is coupled with an unstructured-grid wave model (SWAN) and the unstructured-grid version of the Los Alamos Sea Ice Model (UG-CICE) to form the FVCOM–SWAVE–UG-CICE framework. Using this fully coupled model, simulations were conducted for the Great Lakes. Results of the modeled ice concentration, ice thickness, and significant wave heights were reported and validated against observational data from the U.S. National Ice Center (NIC) and in-situ under-ice measurements. To further study the coupling effects, results of the proposed model were also compared with those from no coupling and one-way coupling (focusing only on ice-induced wave attenuation) models. Comparative analyses demonstrated significant improvements in the predicted ice concentration with the proposed fully coupled model. These findings reveal the importance of incorporating ice–wave interactions in accurately predicting ice cover dynamics in freshwater systems.

1. Introduction

Ice plays a critical role in the ecosystems of both coastal oceans and freshwater lakes, where it regulates surface fluxes (Uhlíková et al., 2023), influences regional precipitation (Kopec et al., 2016), controls coastal erosion (Kostopoulos et al., 2018), and impacts fisheries (Collingsworth et al., 2017). Aquatic ecosystem services, including navigation (Millerd, 2007) and recreation (Chin et al., 2018), are also strongly affected by seasonal ice cycles. Additionally, total ice cover and ice duration can significantly impact the economy and health of coastal communities (Durkalec et al., 2015; Gudmestad, 2020). In the Laurentian Great Lakes, where average ice cover has decreased by nearly 40% over the last 40 years (Cannon et al., 2024), understanding ice dynamics has become increasingly important, particularly during periods when the lakes are not "locked in", *i.e.*, not fully ice-covered. This includes both the shoulder seasons, when ice is forming and melting, as well as peak ice conditions during low-ice years, when total ice cover across all five lakes (Superior, Huron, Michigan, Erie, and Ontario) may not exceed 10%. Ice physics during these periods are especially dynamic and require a multifaceted modeling approach to provide actionable and reliable ice forecasts for coastal management and broader Earth system modeling.

One of the main challenges in such modeling approaches is accounting for wave and ice interactions, particularly along the ice fringe, where surface waves and ice can interact directly. The ice and wave interaction is a complex and two-way coupled process (Squire, 2020, 2022; Dolatshah et al., 2018; Passerotti et al., 2022). On the one hand, the presence of ice significantly impacts wave dynamics through

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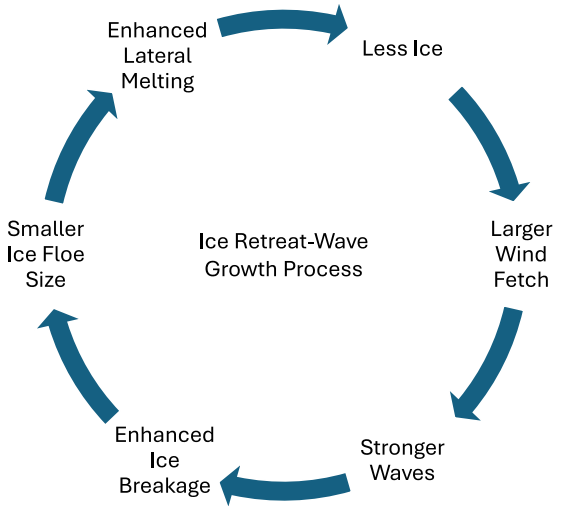


Fig. 1. Ice-retreat wave growth process resulted from wave-induced ice breakage.

attenuation of wave energy (ice-induced wave attenuation). On the other hand, propagating waves can fracture ice into smaller floes (wave-induced ice breakage). A complete model should consider this interaction to accurately reflect the physics of the problem.

The influence of ice on waves, say ice-induced wave attenuation, was incorporated into the Finite-Volume Community Ocean Model (FVCOM) by Bai et al. (2020). This enhancement allowed the model to simulate the attenuation of wave energy by ice, improving the modeling of coastal and offshore wave climates in the Great Lakes. Compared with observation data, Bai et al. (2020) demonstrated that the wave attenuation capability could enhance the model predictions. Despite this advancement, their model did not account for the reverse process: the influence of waves on ice, specifically wave-induced ice breakage.

Zhang et al. (2020) analytically evaluated wave-induced ice breakage and demonstrated that this phenomenon plays a significant role in wave–ice interaction modeling. Through wave-induced ice breakage, waves can fracture ice into smaller floes, leading to a feedback loop where broken ice floes are more susceptible to further melting and breakup, allowing stronger waves to form (Fig. 1). This process, known as ice retreat-wave growth, can significantly influence ice–wave dynamics in cold regions. Therefore, the absence of wave-induced ice breakage in modeling represents a significant gap in the model's ability to fully simulate ice–wave interactions. This limitation hampers the understanding of coupled dynamics between sea ice and ocean waves, which is critical for accurate forecasting and climate modeling, especially in regions experiencing rapid changes in ice cover due to climate change.

This paper addresses this gap by integrating wave-induced ice breakage into modeling, completing the two–way coupling of ice and wave interactions. By introducing this capability, the model can simulate the full spectrum of ice–wave dynamics, providing a more comprehensive tool for studying these interactions. This enhanced model is implemented into the FVCOM and is validated using observational data from the Great Lakes, a region that experiences significant seasonal ice cover and wave activity. The Great Lakes, with their large size and physical processes similar to coastal oceans — such as extensive fetch (up to ~400 km), high waves, and phenomena like pancake and rafted ice — provide an ideal setting for testing and refining the ice–wave models. Additionally, comparative analyses are conducted in this study between different modeling approaches: no coupling, one-way coupled model (considering only ice-induced wave attenuation), and fully coupled model (incorporating both ice-induced wave attenuation and wave-induced ice breakage) as shown in Fig. 2. This approach not only illustrates the improvements brought by the suggested enhancements but also offers insights into the critical feedback mechanisms between ice and waves, enhancing current predictive capabilities and understanding of these processes in the context of a changing climate.

The paper is organized as follows to present a comprehensive investigation into the fully coupled ice–wave interaction model using FVCOM. Section 2 introduces the methodology and data by detailing the model and its implementation in FVCOM through integration of wave and ice modules, including descriptions of the ice-induced wave attenuation and wave-induced ice breakage mechanisms. Data sources and numerical setup for simulations are also outlined in this section. Results and discussions are presented in Section 3, elaborating on findings from simulations using the enhanced FVCOM model. This section emphasizes analyses of ice–wave dynamics and comparisons with observation data using different coupling scenarios. Finally, the main findings are summarized in the Conclusions section.

2. Methodology and data

This section elaborates on the methodology used in this paper and introduces the data sources for modeling and validation.

2.1. FVCOM–SWAVE–UG-CICE model

In this paper, the proposed model will be implemented in the FVCOM–SWAVE–UG-CICE framework to study wave–ice interactions in the Great Lakes region. This framework includes the FVCOM model, which is internally coupled with SWAVE and UG-CICE models.

The Finite-Volume Community Ocean Model (FVCOM), initiated and developed by Chen et al. (2003, 2006, 2012), is a widely used unstructured grid and finite-volume model for simulating and understanding coastal dynamics. In this three-dimensional hydrodynamic model, the domain is divided into small, interconnected, triangular control volumes. These control volumes are composed of unstructured triangular elements in the horizontal domain along with a vertical terrain-following vertical coordinate scheme. Fluxes in these non-overlapping control volumes are computed by solving the governing primitive equations, providing high-resolution simulations of fluid motion, temperature, salinity, and other oceanic properties. FVCOM can utilize a wide variety of input data, such as those for surface meteorology, river discharges, precipitation, and open ocean boundaries. The capability of utilizing multiple set of input data is important as more real-time coastal ocean and atmospheric measurements become available. Moreover, the FVCOM is based on a modular design that allows coupling with the essential model components needed for different applications. The SWAVE and UG-CICE models are some examples of such internally coupled modules.

FVCOM-SWAVE, developed by Qi et al. (2009), is an advanced unstructured-grid finite-volume surface wave model based on the SWAN (Simulating WAves Nearshore) model. The SWAN model was originally created by Booij et al. (1999) and enhanced by the SWAN Team (2006). This third-generation surface wave model simulates shallow-water surface waves by solving the wave action balance equation, accounting for processes like bottom friction, wave–wave interactions, and shallow water wave breaking. While SWAN employs a curvilinear-structured grid with fully implicit finite-difference algorithms, several attempts were made to adapt this model into a finite-element-based unstructured-grid version to address the complexities of irregular coastal geometries (Hsu et al., 2005; Qi et al., 2009). Building on these advancements, FVCOM-SWAVE integrates finite-volume methods with SWAN, resulting in a model that combines the computational efficiency and simplicity of finite-difference methods with the geometric flexibility of finite-element methods (Qi et al., 2009).

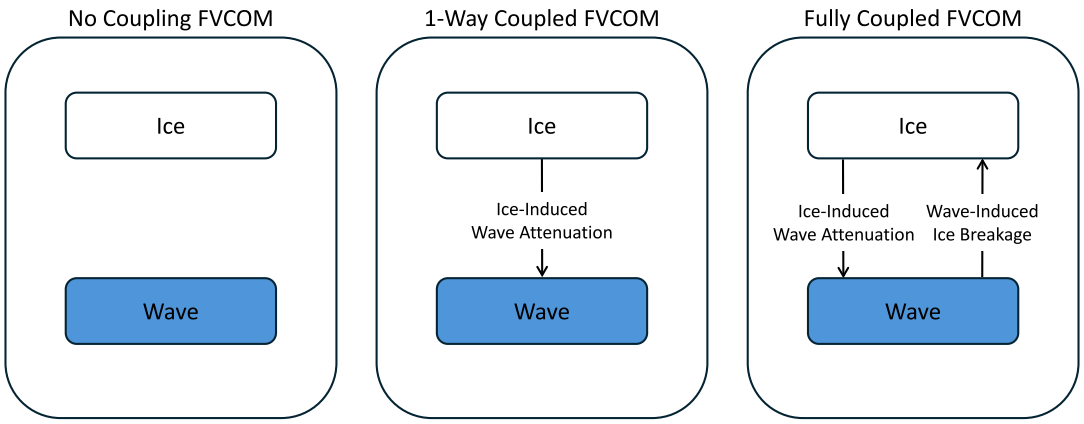


Fig. 2. Coupling schemes evaluated in this paper. The fully coupled FVCOM model is proposed and results are compared with those of 1-way coupled and no coupling models.

Evolved by Gao et al. (2011), UG-CICE is a sophisticated unstructured-grid, finite-volume adaptation of the original Los Alamos Community Ice Code (CICE). The original CICE model utilizes energy-conserving thermodynamic equations (Hunke and Lipscomb, 2006), energy-based ridging schemes of Thorndike et al. (1975), Hibler III (1979), and Lipscomb et al. (2007), with elastic-viscous-plastic ice momentum equations (Hunke and Dukowicz, 1997, 2002; Hunke, 2001) to model sea ice dynamics for structured-grid ocean models. Built on this platform, the UG-CICE model discretizes the governing equations and solves them via flux calculations over non-overlapping unstructured triangular meshes. This finite-volume approach enhances mass conservation within individual control volumes and the entire computational domain. This model effectively simulates two-dimensional fields of ice cover concentration, velocity, and thickness in cold regions such as the Arctic Ocean (Gao et al., 2011; Zhang et al., 2016) and the Great Lakes (Bai et al., 2013; Anderson et al., 2018; Fujisaki-Manome et al., 2020; Bai et al., 2020; Cannon et al., 2023, 2024). Further modifications to enhance the FVCOM model's performance will be discussed in Section 2.5.

While alternative wave–ice interaction models (*e.g.*, Roach et al. (2018), Bateson et al. (2022), Day et al. (2024)) could have been employed, the use of FVCOM in this study provides several notable advantages. First, FVCOM utilizes an unstructured grid, which enables accurate representation of complex coastlines and bathymetries. Second, as previously noted, FVCOM has been successfully applied to modeling the Great Lakes, with its hydrodynamic performance thoroughly validated. Furthermore, incorporation of this model into NOAA's Great Lakes Operational Forecast System underscores its reliability and utility in operational forecasting.

2.2. Ice-induced wave attenuation

Ice-induced wave attenuation refers to the reduction in wave energy and amplitude as waves propagate through and interact with ice. The scheme used in this study is similar to that presented by Bai et al. (2020). As mentioned earlier, the FVCOM-SWAVE model is utilized to solve for the wave dynamics in the computational domain. Originally, the SWAVE model solves the non-linear wave–wave interactions via the wave action balance equation (Qi et al., 2009):

$$\frac{\partial N}{\partial t} + \nabla \cdot \left[\left(\overrightarrow{C_g} + \vec{U} \right) N \right] + \frac{\partial C_\sigma N}{\partial \sigma} + \frac{\partial C_\theta N}{\partial \theta} = \frac{S_{tot}}{\sigma}, \tag{1}$$

where N denotes the wave action density spectrum, t signifies time, $\overrightarrow{C_g}$ represents the wave group velocity, $\vec{U}$ is the velocity vector of the ambient flow, σ and θ refer to the intrinsic (relative) frequency and wave direction, and C_σ and C_θ are the velocities of wave propagation in the spectral space (σ, θ). S_{tot} is the source–sink term and is defined as:

$$S_{tot} = S_{in} + S_{nl} + S_{ds,w} + S_{ds,br} + S_{ds,b}, \tag{2}$$

where S_{in} represents the wind energy input, and S_{nl} denotes the energy transfer resulting from nonlinear interactions among spectral components. Additionally, $S_{ds,w}$, $S_{ds,br}$ and $S_{ds,b}$ terms are whitecapping, wave breaking, and bottom friction components of the wave decay, respectively. Detailed description of these terms can be found in SWAN technical manual (2006).

However, the presence of ice affects the source–sink term in Eq. (2). Ice mitigates the wind energy input (S_{in}) and restrains the wave energy decay due to whitecapping ($S_{ds,w}$) and wave breaking ($S_{ds,br}$). These terms are scaled to reflect their effectiveness only in the open water fraction, blocking them in the ice region (blocking effect). Furthermore, ice introduces a new source of damping due to attenuation. Therefore, the wave propagation governing equation of Eq. (1) under the presence of ice is expressed as:

$$\begin{aligned} &\frac{\partial N}{\partial t} + \nabla \cdot \left[\left(\overrightarrow{C_g} + \vec{U} \right) N \right] + \frac{\partial C_\sigma N}{\partial \sigma} + \frac{\partial C_\theta N}{\partial \theta} \\ &= \frac{(1 - C_{\text{ice}})(S_{\text{in}} + S_{\text{ds,w}} + S_{\text{ds,br}}) + S_{\text{nl}} + S_{\text{ds,b}} + C_{\text{ice}} S_{\text{ice}}}{\sigma}, \end{aligned} \tag{3}$$

where C_{ice} signifies the ice concentration and the term $(1 - C_{ice})$ addresses the blocking effect. In this approach, the source and sink terms are weighed by ice concentration, whereas there are other models that employ more sophisticated weighting factors for these terms (Herman, 2024). S_{ice} is the new wave energy sink term representing ice cover damping and is expressed as:

$$S_{ice} = -\alpha E C_g, \tag{4}$$

with α and E being the attenuation coefficient and wave spectral density, respectively. Bai et al. (2020) studied several empirical schemes used in the IC4 of the WAVEWATCH III (2016) model for ice attenuation coefficient (IC4M1–IC4M7). The IC4M6 scheme (Rogers et al., 2016) is chosen in this study, as it approximately represents an average of the IC4M1–IC4M7 schemes (Bai et al., 2020). The ice attenuation coefficient from this scheme is a piecewise function expressed below:

$$\alpha = \begin{cases} 2 \times 2.94 \times 10^{-6} & \text{if } f < 0.1 \\ 2 \times 4.27 \times 10^{-6} & \text{if } 0.1 \le f < 0.15 \\ 2 \times 7.95 \times 10^{-6} & \text{if } 0.15 \le f < 0.20 \\ 2 \times 2.95 \times 10^{-5} & \text{if } 0.20 \le f < 0.25 \\ 2 \times 1.12 \times 10^{-4} & \text{if } 0.25 \le f < 0.30 \\ 2 \times 2.74 \times 10^{-4} & \text{if } 0.30 \le f < 0.35 \\ 2 \times 4.95 \times 10^{-4} & \text{if } 0.35 \le f < 0.40 \\ 2 \times 8.94 \times 10^{-4} & \text{if } f \ge 0.40, \end{cases} \tag{5}$$

where f is the wave frequency in Hz (see Fig. 3). While the wave attenuation model used in this study relies solely on wave frequency, more advanced models incorporate ice parameters such as thickness and floe size into the parameterization (De Santi et al., 2018; Yu et al.,

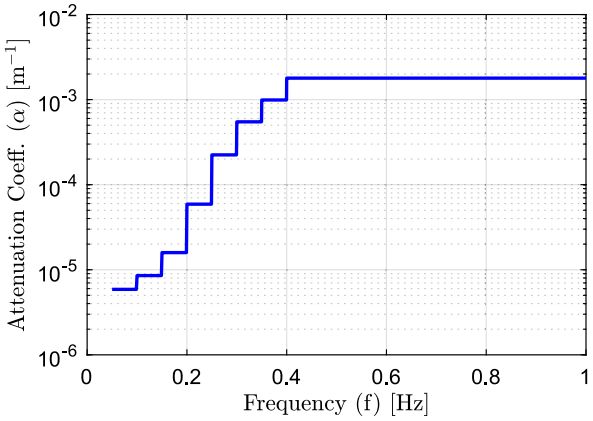


Fig. 3. Attenuation coefficient (α) against the wave frequency (f) based on the IC4M6 scheme (Rogers et al., 2016). Results are shown in logarithmic scale for better visualization. The attenuation coefficient stays constant for frequencies higher than 1 Hz.

2022). Future implementation of these advanced models in coupled frameworks could enhance predictive accuracy. It is important to note that hereafter, the term "ice-induced wave attenuation" in this paper accounts for both the blocking effect and ice damping.

In a nutshell, to account for ice-induced wave attenuation in the FVCOM–SWAVE–UG-ICE model, the ice attenuation coefficient, α, for the wave is first calculated using Eq. (5) in each time step. The wave energy sink term due to ice damping, S_{ice}, is then calculated using Eq. (4). With the ice concentration from the UG-CICE module, the wave spectra evolution is solved for the control volumes using Eq. (3) in the FVCOM-SWAVE module.

2.3. Wave-Induced Ice Breakage

The Wave-Induced Ice Breakage refers to the fragmentation of sea/lake ice into smaller pieces due to mechanical forces exerted by waves. Accounting for this phenomenon in the proposed model involves extensive modifications to the UG-CICE module. The process consists of three main steps: breaking criterion, new representative floe diameter, and floe size-dependent lateral melt. These steps are elaborated in the following subsections.

2.3.1. Breaking criterion

The first step in implementing Wave-Induced Ice Breakage is to check whether the criterion for ice breakage is met. This is to determine whether the incoming waves have enough energy to overcome the ice's critical significant strain. As suggested by Williams et al. (2013), passing waves can break ice if the significant strain amplitude, E_s, is larger than the critical significant strain amplitude, E_c. This breaking criterion is given by Williams et al. (2013) as:

$$E_s > E_c \,. \tag{6}$$

The critical significant strain amplitude, E_c, can be calculated as:

$$E_c = \epsilon_c \sqrt{\frac{-2}{\log \mathbb{P}_c}} \,, \tag{7}$$

where $\mathbb{P}_c = e^{-1} \approx 0.37$ is the approximate value for the probability criterion, and ϵ_c is the critical breaking strain defined as

$$\epsilon_c = \frac{\sigma_c}{Y^*} \,. \tag{8}$$

The terms σ_c and Y^* are the flexural strength and effective Young's modulus, respectively, defined as:

$$\sigma_c = \sigma_0 \exp\left(-5.88\sqrt{\upsilon_b}\right) \,, \tag{9}$$

$$Y^* = Y_0(1 - 3.51\upsilon_b) - 1 \text{ GPa} \,, \tag{10}$$

where υ_b is the brine volume fraction, while $\upsilon_b = 0$, σ_0= 1.76 MPa, and $Y_0 \approx 10$ GPa are the approximate constant values for freshwater ice (Timco and O.'Brien, 1994; Williams et al., 2013).

The significant strain amplitude, E_s, in Eq. (6) can be calculated as following

$$E_s = 2\sqrt{\int_0^\infty S(\omega)E^2(\omega)d\omega} \,, \tag{11}$$

where $S(\omega)$ is the spectral density distribution of wave energy, and ω is the angular frequency. The strain amplitude, $E(\omega)$, is characterized as:

$$E(\omega) = \frac{h_{\text{ice}}}{2} k_{\text{ice}}^2 W(\omega) \,, \tag{12}$$

where h_{ice} is the ice thickness, and k_{ice} is the wave number for the ice-covered ocean (see Eq. A.7 of Williams et al., 2013). Finally, $W(\omega)$ is expressed as:

$$W(\omega) \approx \frac{k_{\text{ice}}|\mathcal{T}|}{k} \,, \tag{13}$$

where $\mathcal{T}$ is the transmission coefficient for waves traveling from open water into ice-covered water (Williams and Porter, 2009), and k is the deep water propagation wave number ($k = \omega^2/g$). It is noteworthy that some studies, such as Collins et al. (2018) and Yu et al. (2022), have indicated that, at certain frequencies, the wave number in ice exhibits minimal deviation from its open-water value ($k_{\text{ice}} \simeq k$). While the model proposed in this paper is not limited to this assumption, assuming $k_{\text{ice}} = k$ can offer significant reductions in computational time during simulations. Additionally, alternative parameters, such as the ice break-up threshold reported by Voermans et al. (2020) could be employed in place of the breaking criterion discussed in this study.

In summary, to check the wave breaking criterion, the critical significant strain amplitude, E_c, is first calculated through Eqs. (7)–(10). Subsequently, the significant strain amplitude, E_s, is computed using Eqs. (11)–(13). If the breaking criterion of $E_s > E_c$ is met, wave-induced ice breakage has occurred. Therefore, the steps in Sections 2.3.2 and 2.3.3 are also considered. If the breaking criterion is not satisfied, wave-induced ice breakage has not occurred in that cell, and the computation proceeds with the original ice floe diameter. It should be noted that the data required for these steps come from both the UG-CICE and SWAVE modules.

2.3.2. New representative floe diameter

It was mentioned that wave-induced ice breakage occurs when the breaking criterion is satisfied. The original implementation of UG-CICE model by Gao et al. (2011) considers a relatively large value for floe-size diameter, D_{max} = 300 m, following Steele (1992). After ice breakage, the ice floe is divided into smaller pieces so it is crucial to define a representative floe diameter, $D(i,j)$, to proceed with computations in the cell (i,j). This representative floe diameter is suggested by Bennetts et al. (2017) as:

$$D(i,j) = [1 - a_{bk}(i,j)]D_0(i,j) + a_{bk}(i,j)D_{bk}(i,j) \,, \tag{14}$$

where $D_0(i,j)$ is the representative diameter in the cell at the beginning of the time step (old representative diameter), and $a_{bk}(i,j) = L_{bk}/L_{cl}$ is the weighting coefficient, with L_{bk} as the traveling distance of the wave spectrum and L_{cl} as the length of the grid cell. The term D_{bk} denotes the broken floe diameter and is expressed by Bennetts et al. (2017) following a split power law, $p(d)$, as:

$$D_{bk} = \int_{D_{mn}}^\infty p(d)d \; \mathrm{d}d = \frac{\mathbb{P}_0\gamma_0\beta_0\left(D_{mn}^{1-\gamma_0} - D_{cr}^{1-\gamma_0}\right)}{\gamma_0 - 1} + \frac{(1-\mathbb{P}_0)\gamma_1\beta_1 D_{cr}^{1-\gamma_1}}{\gamma_1 - 1} \,, \tag{15}$$

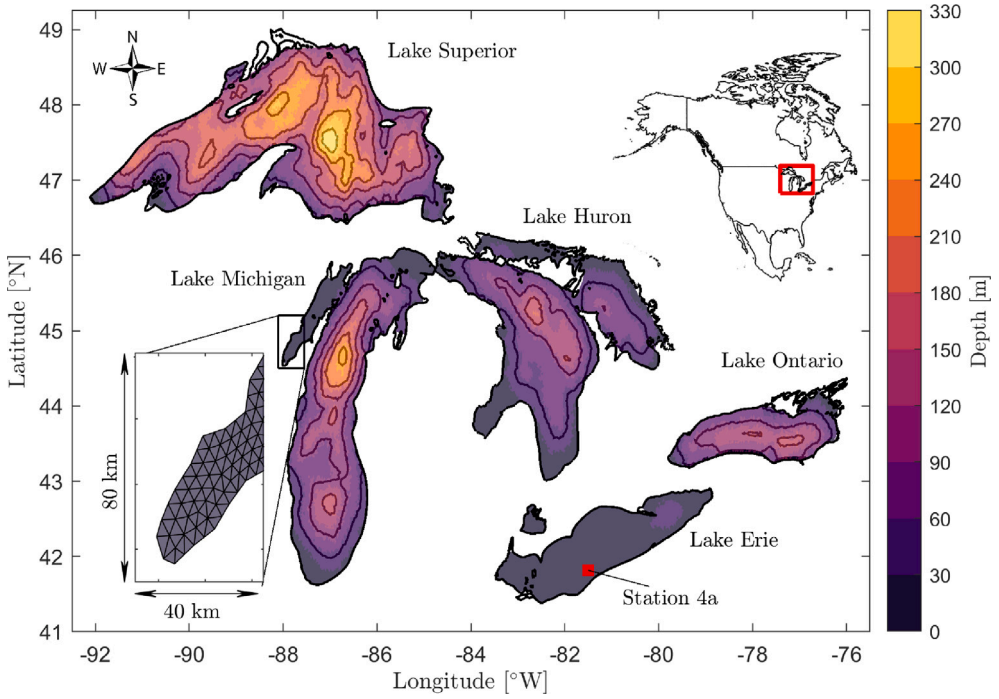


Fig. 4. The Great Lakes bathymetry and position in the North America. A portion of the northwestern side of Lake Michigan (Green Bay) is magnified to show the unstructured meshes utilized in the FVCOM computations.

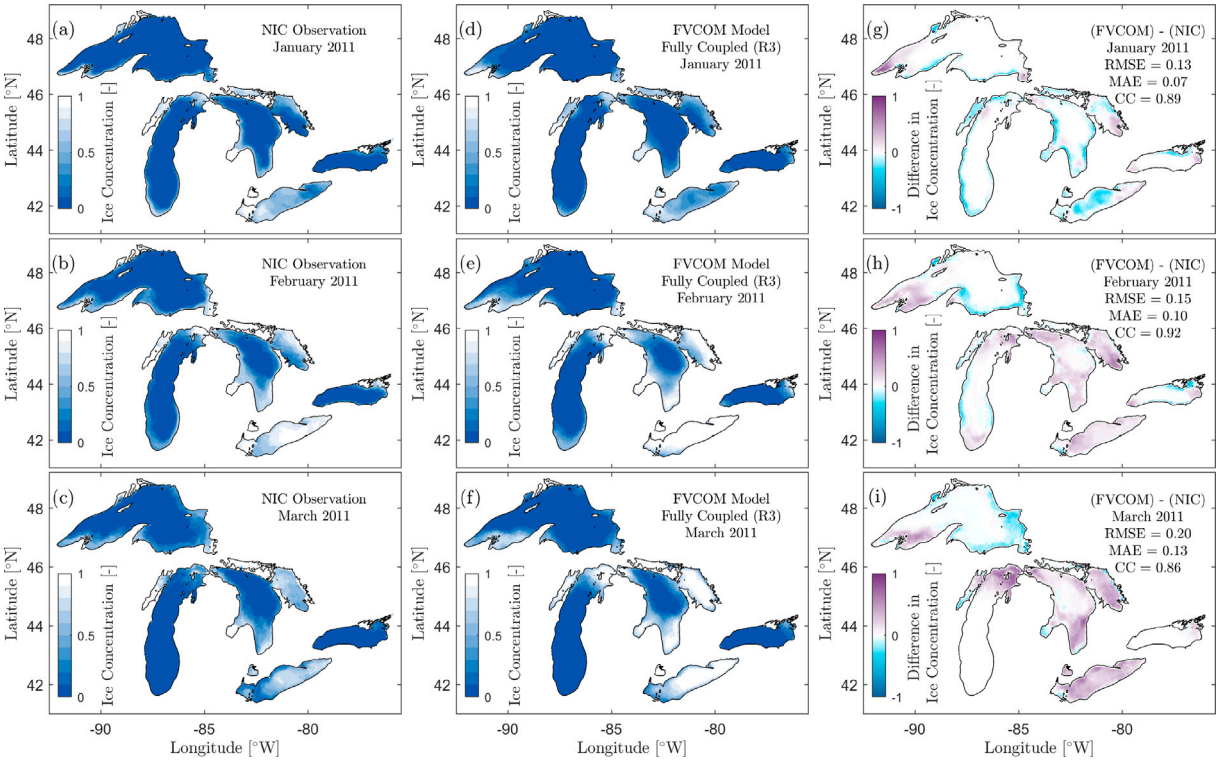


Fig. 5. Monthly average ice concentration in winter 2011 from fully coupled FVCOM model (R3) compared with the NIC observation data. NIC satellite measurements (a–c) are compared with model predictions (d–f) for different months. The difference between FVCOM simulation results and NIC observations for each month is also plotted (g–i). These plots include the relative error metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Correlation Coefficient (CC).

where d is the floe diameter, D_{mn} is the minimum floe diameter, and D_{cr} is the critical diameter marking the transition from small to large floes, shown to range from 15–40 m by Toyota et al. (2011). Parameters γ_0 and γ_1 are determined by fitting a function to observations (see Toyota et al., 2011). In this study, values $\gamma_0 = 1.15$ and $\gamma_1 = 2.15$ are used. Also, Bennetts et al. (2017) suggested the following relations for β_0 and β_1:

$$\beta_0 = \left(D_{mn}^{-\gamma_0} - D_{cr}^{-\gamma_0}\right)^{-1} , \tag{16}$$

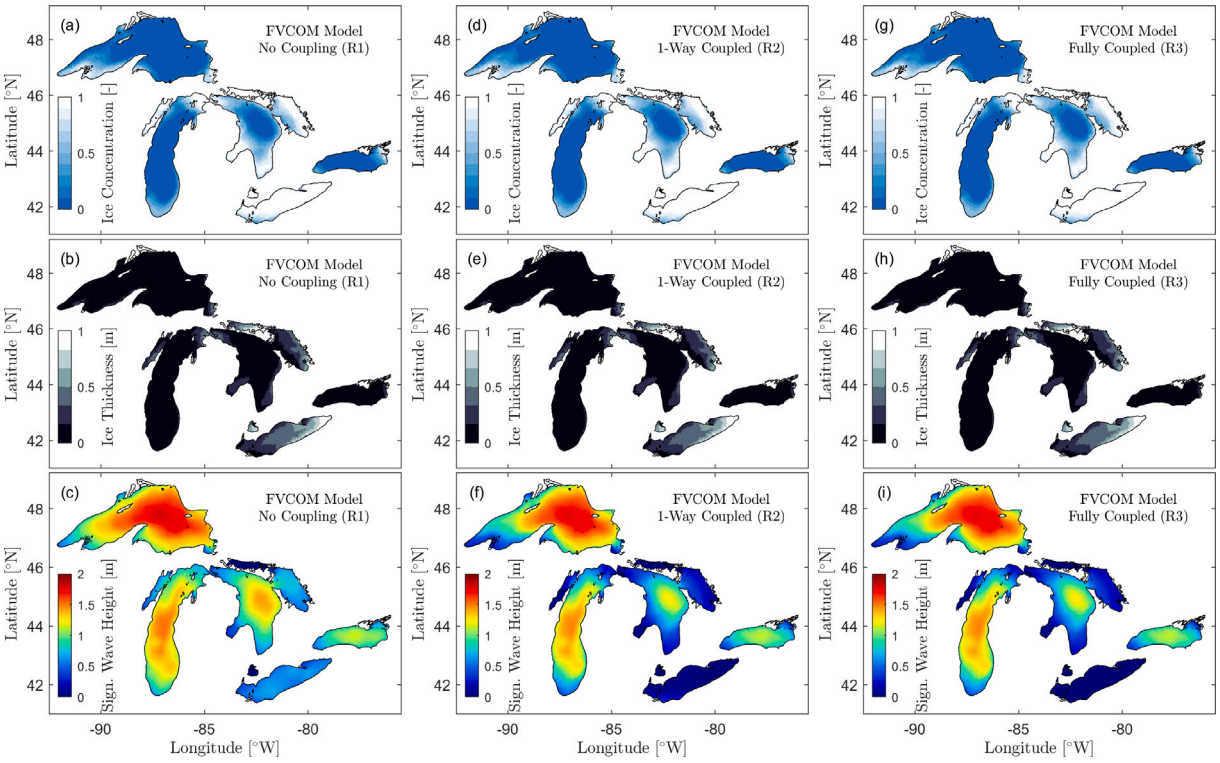


Fig. 6. Monthly average data from No Coupling Model (a–c), 1-Way Coupled Model (d–f), and Fully Coupled Model (g–i). Results are shown for ice concentration (a, d, and g), ice thickness (b, e, and h), and significant wave height (c, f, and i) in February 2011.

$$\beta_1 = D_{cr}^{\gamma_1} \, . \tag{17}$$

Additionally, $\mathbb{P}_0 \in [0, 1]$ weights the distribution towards small floes (large $\mathbb{P}_0$) or large floes (small $\mathbb{P}_0$) and can be defined as:

$$\mathbb{P}_0 = 1 - q \left(\frac{D_{pr}}{D_{cr}} \right)^{\gamma_1} \, , \tag{18}$$

where $D_{pr} = \lambda/2$ is the predicted breakup diameter, directly related to the dominant wavelength λ for the wave spectrum. The q denotes the portion of the floe diameters that are greater than D_{pr}.

In short, once the ice breaking criterion is met (Section 2.3.1), wave-induced ice breakage occurs, and a new representative floe diameter needs to be calculated for each cell. To achieve this, the broken floe diameter, D_{bk}, is first calculated from Eqs. (15)–(18). The representative floe diameter for each cell at each time step is then computed by Eq. (14). This process is applied at the beginning of the UG-CICE module and is repeated for each cell at each time step as long as the breaking criterion is met. This ensures that in case of ice breakage, the rest of the computations in the UG-CICE module are conducted on the new reduced representative floe diameter. It should be noted that the version of the UG-CICE utilized in this study does not account for the growth of the floe size through consolidation and/or lateral growth of floe in the freezing period. This capability is available in some of the newer ice models through sub-grid scale floe-size distribution (FSD).

2.3.3. Floe size dependent lateral melt

Introducing the breaking criterion, as discussed earlier, wave-induced ice breakage can result in smaller floe diameters. Reducing the representative floe diameter increases the lateral area of ice floes, thereby enhancing lateral ice melting. According to Steele (1992), this can be addressed by adjusting the fraction of ice that melts laterally:

$$r_{\text{lat}} = \frac{\pi \Delta t w_{\text{lat}}}{\mu D} \, , \tag{19}$$

where Δt is the time step, $\mu = 0.66$ is the geometric parameter, and D is the new representative floe diameter. Also, w_{lat} is lateral melt rate and can be derived from:

$$w_{\text{lat}} = m_1 \Delta T^{m_2} \, , \tag{20}$$

where $m_1 = 8 \times 10^{-6}$ is the calibrated lateral melting coefficient for the freshwater ice, ΔT is the temperature difference between the water surface and the ice bottom, and $m_2 = 1.36$ is the empirical constant suggested by Maykut and Perovich (1987). It is worth mentioning that another wave–ice interaction process is part of the wave radiation stress acting onto the lateral side of the ice floe. This process, which contributes to the advection is not considered in this study.

2.4. Data sources

Different data sources are used in this paper mainly for two reasons: (1) as input (forcing files) into the FVCOM–SWAVE–UG-CICE model, and (2) to assess and validate the simulation results. These sources are detailed below.

The model was driven by meteorological data sourced from the North American Regional Reanalysis (NARR) project (Mesinger et al., 2006), with a temporal resolution of 3 h and a spatial resolution of 32 km. Compared to similar hindcast reanalysis products (such as ERA5 and NCAR), the winter air temperature data from NARR are found to be less biased for the Great Lakes (Cannon et al., 2024). This is crucial as air temperature significantly influences lake ice cover. However, direct comparisons between National Data Buoy Center (NDBC) and NARR forcing wind data revealed that NARR data generally underestimates lake surface winds. Bai et al. (2020) showed that the average difference in wind speed of the NARR data is 15%–20% lower than the NDBC measured data. Since the accurate wind forcing data is a pivotal component for SWAVE wave modeling (Mao et al., 2016), a 20% boost on NARR wind data was considered to improved modeling accuracy. Furthermore, water surface temperature from Great Lakes Surface Environmental Analysis (GLSEA) were used for ice initialization of the model (see Section 2.5).

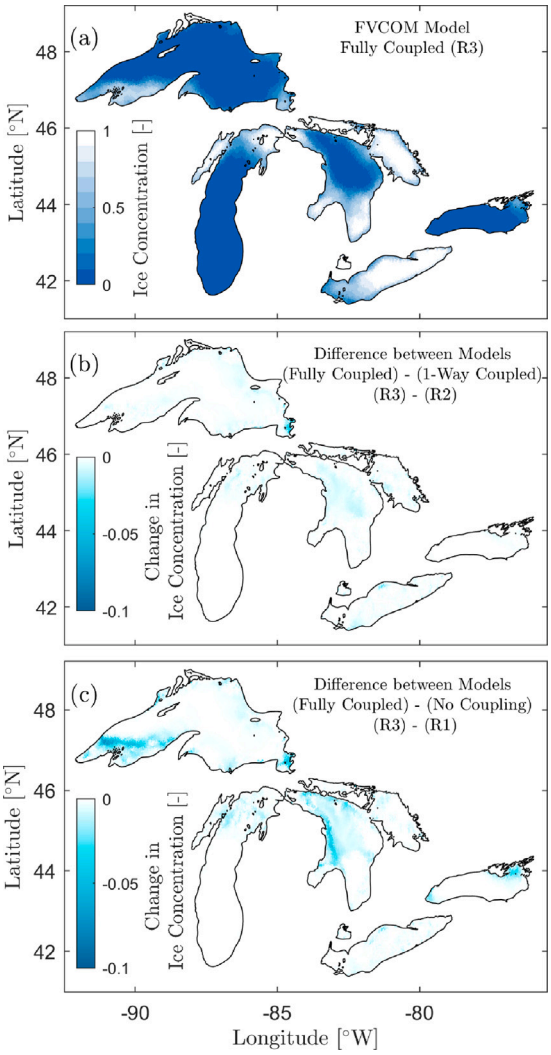


Fig. 7. Effect of coupling on monthly averaged ice concentration data in March 2011. Results are shown for (a) fully coupled model, (b) difference between fully coupled model and 1-way coupled model, and (c) difference between fully coupled model and no coupling model.

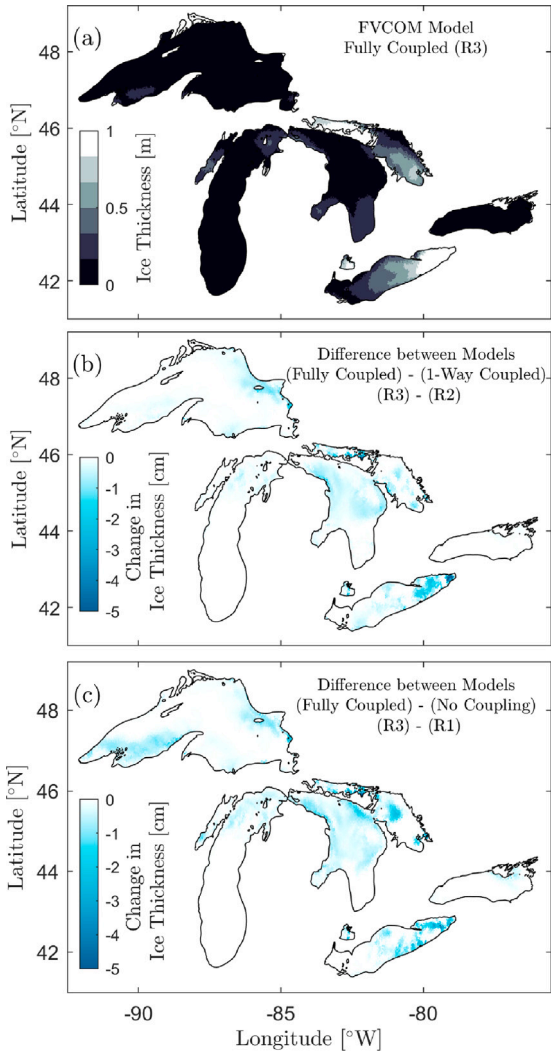


Fig. 8. Effect of coupling on monthly averaged ice thickness data in March 2011. Results are shown for (a) fully coupled model, (b) difference between fully coupled model and 1-way coupled model, and (c) difference between fully coupled model and no coupling model.

For validation purposes, satellite-retrieved ice concentration data from Moderate Resolution Imaging Spectroradiometer (MODIS) images are used. These data, processed by the National Ice Center (NIC) and provided by National Oceanic and Atmospheric Administration Great Lakes Environmental Research Laboratory (NOAA GLERL), help evaluate the ice modeling performance. Additionally, wave and ice data from Shallow Water Ice Profiler (SWIP) measurements in Lake Erie (conducted in winter 2011 at station 4a, see Fig. 4) are utilized (Hawley et al., 2018).

2.5. Numerical setup

The FVCOM–SWAVE–UG-CICE model used in this study incorporates fully coupled interactions between waves and ice across the Great Lakes region, including Lake Superior, Lake Michigan, Lake Huron, Lake Erie, and Lake Ontario (Fig. 4). Water exchange is allowed only between Lake Michigan and Lake Huron, with other lakes isolated. Unstructured grids were generated to achieve an average horizontal resolution of approximately 3.5 km. In the vertical domain, bathymetry data with approximately 90 m cell size (on the water surface) from NOAA were utilized to create 21 terrain-following vertical layers. These vertical layers, also known as sigma layers, have an increased resolution at the water surface and at the bottom to better track the boundary layer processes.

Compared to the original version, the FVCOM model used in this study is the Great Lakes (GL) FVCOM. The GL-FVCOM model benefits from an improved stability by implementing a second-order leapfrog scheme for the time integration (Wang et al., 2023). This improvement is accomplished by an explicit and special handling of the Coriolis parameter to minimize its associated phase error, yielding to a more stratified vertical temperature structure. A time step of 60 s was set for the internal mode with a splitting ratio of 20 for the external mode (3 s). Also, a time step of 60 s was considered for the ice and wave module operations. Additionally, basal and lateral melting coefficients (Cannon et al., 2023) and ice strength values in the UG-CICE module are calibrated for better simulation of the Great Lakes freshwater physics and thermodynamics. In this study, basal and lateral melting coefficients of $m_1 = 0.03$ [–] and $m_2 = 8 \times 10^{-6}$ [m/(s °C$^{1.36}$)] were employed, as verified for the Great Lakes by Cannon et al. (2023). Also, ice strength of $P^* = 5$ [kN/m^2] were considered based on the sensitivity analysis (see Fig. 18 in Appendix A). Moreover, the model is modified to account for ice initialization to represent more accurate initial conditions. This is accomplished using various set of data including US NIC ice observation data and GLSEA water surface temperature

Table 1
Numerical setup for the simulation runs (see Fig. 2 for more details on coupling schemes)

Run	Coupling	Description
R1	No coupling	No coupling between wave and ice modules
R2	1-Way coupled	Ice-Induced Wave Attenuation only
R3	Fully coupled	Ice-Induced Wave Attenuation and Wave-Induced Ice Breakage
R4	1-Way coupled	Ice-Induced Wave Attenuation only - regenerated from (Bai et al., 2020) model

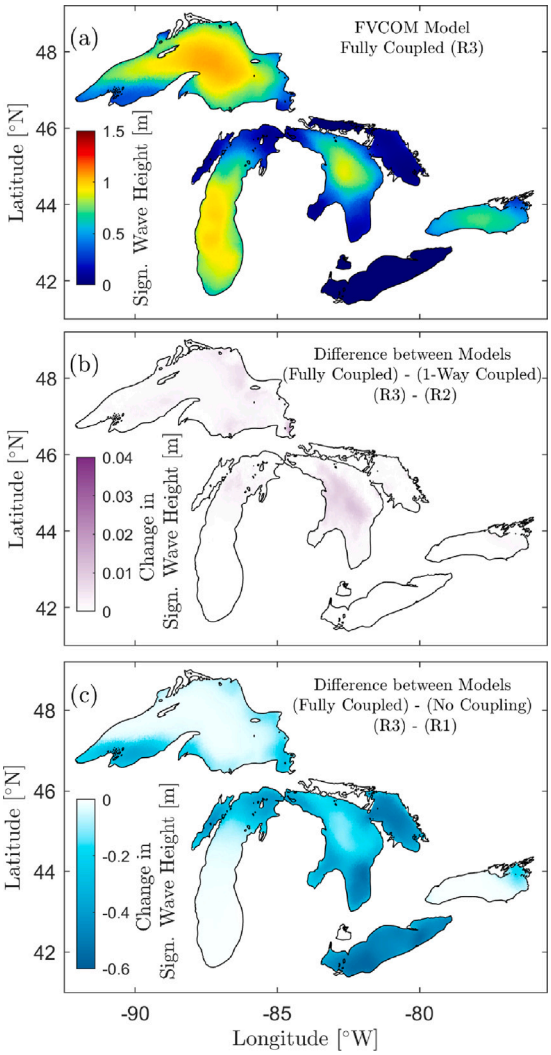


Fig. 9. Effect of coupling on monthly averaged significant wave height data in March 2011. Results are shown for (a) fully coupled model, (b) difference between fully coupled model and 1-way coupled model, and (c) difference between fully coupled model and no coupling model.

(linear transition from GLSEA temperature at the water surface to 4 °C at bed). These modifications improved the model performance and yielded to more accurate simulation results (see Section 3 and Appendix A for the results).

Table 1 presents details of the model runs conducted, focusing on the winter of 2011 to align with Hawley et al. (2018) SWIP experimental measurements for ice data.

3. Results and discussion

This section presents the simulation model results, along with validation, comparisons, and discussions.

3.1. Ice and wave modeling

Fig. 5 displays a comparative analysis of ice concentration results from NIC observations 5(a)–5(c) and the fully coupled model 5(d)–5(f) for winter 2011. Figs. 5(g)–5(i) illustrate the difference between simulation results and NIC observations for each month along with relative error metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Correlation Coefficient (CC). NIC observations serve as a benchmark for model validation. The fully coupled model, incorporating ice-induced wave attenuation and wave-induced ice breakage, shows a high degree of correlation with the observed data across all three months, indicating its robustness in simulating real-world conditions. The model also captures the trend of freezing in different lakes effectively. Discrepancies are observed in Lake Erie (81.2°W, 41.7°N) and Georgian Bay (81.3°W, 45.2°N), where freezing occurs earlier due to shallow water. Although the model simulates the freezing period well 5(g)–5(h), the melting process for these regions is slightly delayed compared to observations 5(i). The delayed melting process results in overestimation of ice in Lake Huron and northern Lake Michigan, increasing the RMSE value in March compared to the previous months. This can be attributed to the lack of streamflow in this model, since no inflow currents from the rivers are considered in this study. Moreover, NARR forcing temperature data in the Great Lakes have been shown to be slightly cold biased in spring that can potentially hinder the melting process in some regions (Cannon et al., 2024). Yet, the fully coupled model's predictions align closely with observations in other regions, highlighting its superior performance. It should be noted that while the US-NIC products are widely used, these data may have uncertainties due to satellite limitations, such as cloud interference and sensor resolution. Moreover, the temporal and spatial averaging in the data processing may also affect the regional accuracy of the US-NIC results.

Fig. 6 summarizes the results of the three main FVCOM models: no coupling model 6(a)–6(c), 1-way coupled model 6(d)–6(f), and fully coupled model 6(g)–6(i). Averaged ice concentration, ice thickness, and significant wave height are reported for February 2011. As mentioned earlier, ice concentration is higher in Lake Erie due to its shallow water. Higher significant wave heights are observed at Lake Superior, where large fetch and depth generate stronger waves. At first glance, all three models show similar ice prediction performance. Ice concentration (6(a), 6(d), and 6(g)) and ice thickness results (6(b), 6(e), and 6(h)) are nearly identical across models, indicating minimal ice breakage during this period. However, the ice-induced wave attenuation effect is evident. For instance, the no coupling model shows a relatively high significant wave height (~0.7 m) in Lake Erie 6(c), despite extensive ice coverage. In reality, the existence of ice inhibits the wave formation and thus, no wave is expected to be generated under solid ice cover. This mismatch stems from the fact that the ice and wave modules in the no coupling model are not coupled and the computations in these modules are conducted separately from each other (*i.e.*, the wave model runs without any information of ice conditions form the ice model). Implementing ice-induced wave attenuation in the 1-way coupled model 6(f) and fully coupled model 6(i) accounts for this and predicts negligible wave generation in Lake Erie. Similar behavior is seen in other high ice concentration areas, such as Georgian Bay. The coupling effects are further explored in the following section.

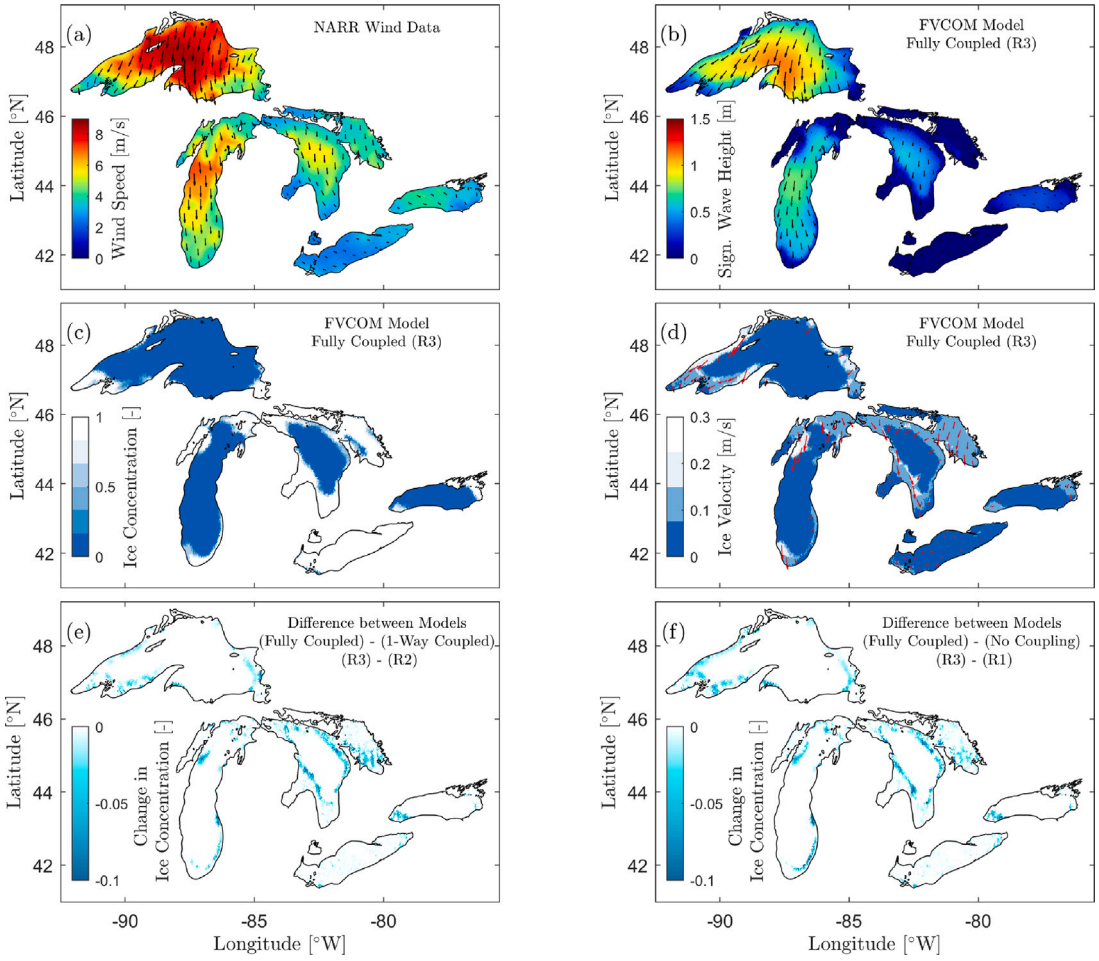


Fig. 10. Daily average data in January 30th, 2011. Results are shown for (a) NARR wind data, (b) Significant Wave Height, (c) ice concentration, (d) ice velocity, (e) ice concentration difference between fully coupled model and 1-way coupled model, and (f) ice concentration difference between fully coupled model and no coupling model. Arrows plotted in (a), (b) and (d) represent the directions of wind, wave and ice velocity, respectively.

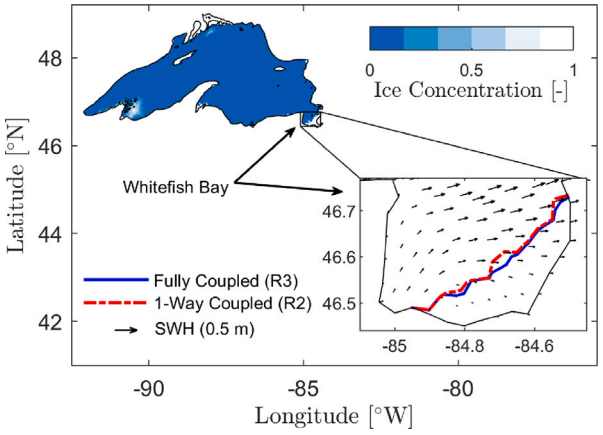


Fig. 11. Ice concentration in the Lake Superior and the effect of coupling in Whitefish Bay (magnified figure) at 12 pm on April 3rd, 2011. The lines in the magnified figure show the edge of the region with ice concentration of 0.8 from fully coupled and 1-way coupled models. While the ice concentration contours are not plotted in the magnified figure for sake of clarity, the arrows represent the magnitude and direction of the significant wave heights.

3.2. Effects of coupling

3.2.1. Overall effects

Fig. 7 illustrates the effects of coupling on monthly averaged ice concentration in March 2011. The fully coupled model's ice concentration 7(a) and its differences with 1-way coupled 7(b) and no coupling models 7(c) are shown. The small difference between the fully coupled and 1-way coupled models 7(b) indicates that the ice breakage phenomenon is not significant during this period. Negative changes suggest that wave-induced ice breakage slightly reduces the average ice concentration, recorded as 4% and 6% compared with 1-way coupled and no coupling models, respectively. These changes are more noticeable along the marginal ice zones (MIZ), like the west side of Lake Michigan and Lake Superior, where ice–wave interactions are stronger. In general, regions with higher fetch, such as the Bering Sea or larger marginal ice zones, are expected to exhibit stronger swell waves and more pronounced ice breakage effects.

The comparison for ice thickness is shown in Fig. 8. It reveals a small reduction in ice thickness across locations, attributed to ice breakage fragmenting ice floes into smaller pieces with reduced thickness. Changes are approximately 2–3 cm, slightly greater for the no coupling model 8(c) compared to the 1-way coupled model 8(b).

Fig. 9 shows coupling effects on predicted significant wave heights in March 2011. Again, averaged values from the fully coupled model 9(a) and differences with the 1-way coupled model 9(b) and no coupling model 9(c) are presented. The fully coupled model shows a slight

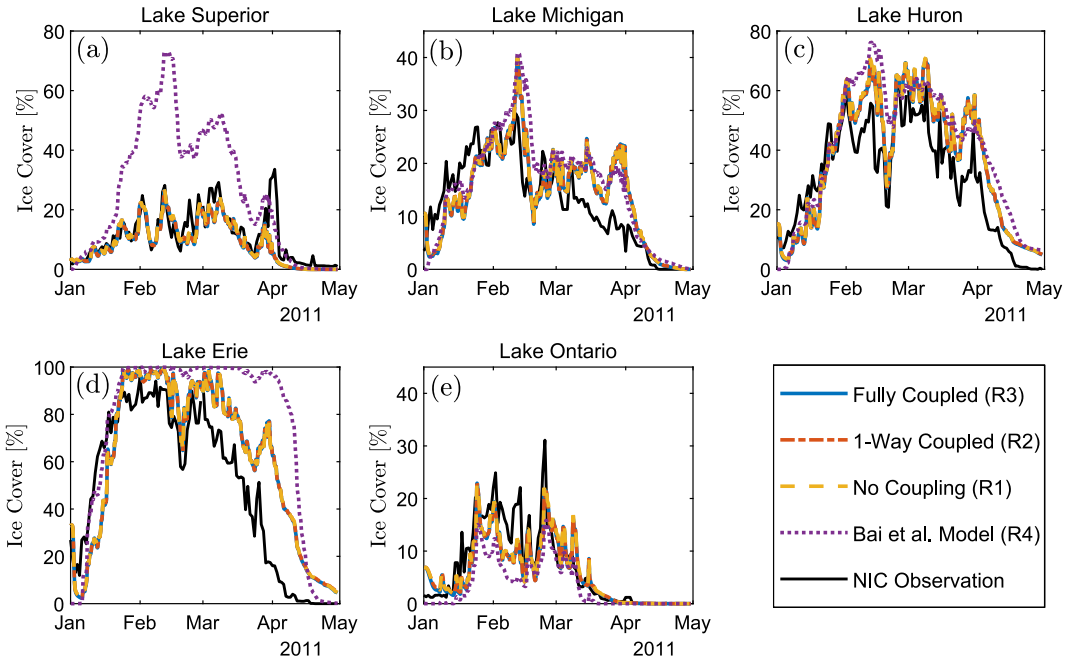


Fig. 12. Daily lake-averaged ice cover in (a) Lake Superior, (b) Lake Michigan, (c) Lake Huron, (d) Lake Erie, and (e) Lake Ontario, in the period of January-May 2011. Predictions from different FVCOM models are compared with the NIC observations. Results of the Model R4 is regenerated from the model presented by Bai et al. (2020).

increase in significant wave heights in some regions compared to the 1-way coupled model. Although changes are small (~2 cm), they correlate with reduced ice concentration due to ice breakage (see Fig. 7(b)). Lower ice concentration reduces ice-induced wave attenuation, increasing significant wave heights. The difference between fully coupled and no coupling models is negative, showing a reduction in wave heights. The no coupling model's wave module lacks feedback from the ice module, falsely generating waves with high significant wave heights in frozen regions. The fully coupled model, with ice–wave feedback, predicts negligible waves in these areas, resulting in significant negative values (Fig. 9(c)), especially in regions with more ice cover. Lake Erie's significant wave height reduction in March reaches ~0.6 m (this reduction was ~0.7 m in February, as discussed under the discussion of Fig. 6).

Reviewing ice concentration (or ice thickness) results, one can note that Figs. 7(b) and 7(c) (or Figs. 8(b) and 8(c)) are not identical. This infers that even though the ice module is identical for the 1-way coupled and no coupling schemes, slightly different ice results can be observed between these two models. This difference stems from the FVCOM module that is coupled with the wave module. As discussed by Niu and Xia (2017), the coupling between FVCOM hydrodynamics and SWAVE module allows wave-induced air-lake momentum and heat transfers influence the lake thermal structures and temperature. These parameters, driven from FVCOM, are among the main inputs for the ice module. Thus, any changes in these parameters may vary the outcome of the ice module. In other words, although the wave module is not directly coupled with the ice module in the 1-way coupled or no coupling models, different generated waves in these modules (see Figs. 6(c) and 6(f)) can indirectly influence the ice results by altering lake thermal structures through the FVCOM module. However, this indirect influence is not tangible and therefore, as discussed earlier, a direct coupling between ice and wave modules is necessary to accurately model the physics of the lakes.

3.2.2. Instantaneous and localized effects

It is worth mentioning that the ice breakage effects are mainly observed at the edges of the ice floes, where the ice–wave interaction occurs and propagating waves can break the ice floes into smaller pieces. Since the shape and location of ice floes in the lakes differ with time, it is crucial to have a closer look at the instantaneous breakage effects (in a shorter time period) other than the monthly average results. To this end, daily average results in January 30th, 2011, are plotted in Fig. 10. The wind speed and wind directions (shown by arrows) are shown in Fig. 10(a). It can be seen that at this date, wind speeds of up to 8.6 m/s are recorded in Lake Superior. This relatively high wind speed and large fetch distances in this lake result in higher significant wave heights, as shown in Fig. 10(b). As was expected, wave directions, indicated by arrows, align with wind directions. Ice-induced wave attenuation results in zero significant wave heights in regions like Lake Erie (Fig. 10(c)). Despite tangible wind speeds, ice presence inhibits wave formation. In non-frozen lakes, wind moves ice floes at up to 0.3 m/s (Fig. 10(d)). Moreover, it can be seen that the ice drift angle is deviated from the wind direction. This is due to several factors such as air- and water-ice drag, internal ice stress, Coriolis force, and coastal friction (Sinsabvarodom et al., 2024; Lukovich et al., 2014; Leppäranta, 2011). It should be noted that the air–ice drag coefficient in this study is calculated based on the constant roughness length and atmosphere stability corrections, while the air–water drag coefficient is calculated based on Large and Pond (1981). Figs. 10(e) and 10(f) show coupling effects. Fig. 10(e) depicts the difference between the fully coupled model and the 1-way coupled model, while Fig. 10(f) shows the difference between the fully coupled model and the no coupling model. Comparing these figures with ice concentration results (Fig. 10(c)), ice breakage effects are evident at ice floe edges, where ice–wave interaction is significant. In this case, incoming waves can break ice and reduce the ice concentration up to 10%. The reduction in ice concentration in the fully coupled model compared to the no coupling model (Fig. 10(f)) is marginally higher than the difference between the fully coupled model and the 1-way coupled model (Fig. 10(e)). As elaborated earlier, this is an indirect influence of the waves due to the coupling between FVCOM and wave modules.

To further study the coupling effect in a shorter period of time, Fig. 11 compares the localized results of fully coupled and 1-way coupled models in Whitefish Bay at 12 pm on April 3rd, 2011. As shown in the figure, a small portion of the south east side of the Lake Superior, also known as Whitefish Bay, is covered with ice at this time. The lines in the magnified figure portray the edge of the region with ice concentration of 0.8 from fully coupled and 1-way coupled models. It

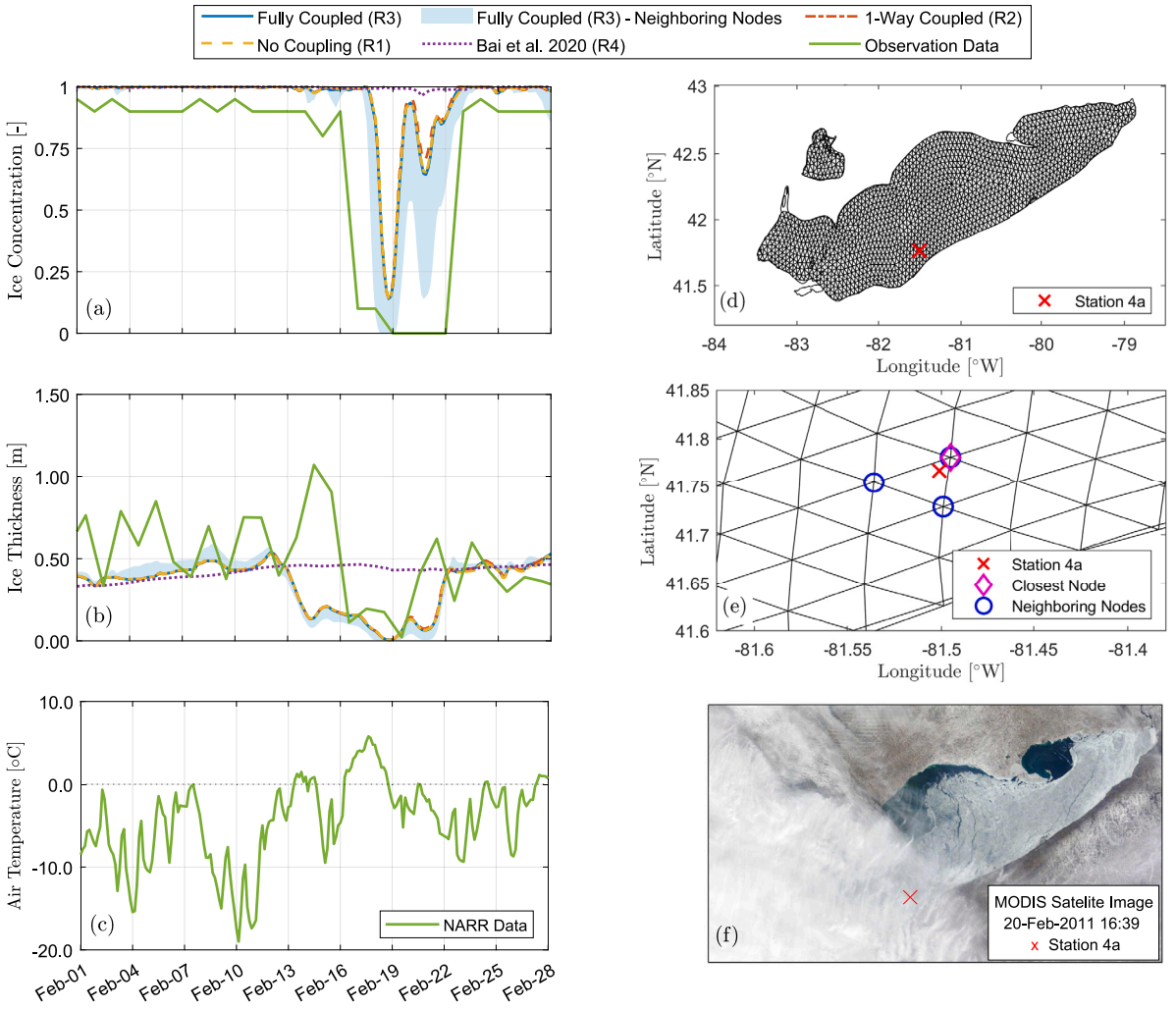


Fig. 13. Comparison between FVCOM results with observations in Station 4a of Lake Erie. Results show comparisons between FVCOM results and (a) NIC ice concentration, and (b) (Hawley et al., 2018) SWIP ice thickness measurements. Air temperature data from NARR is also plotted for the reference (c). Additionally, the plots are shown for (d) Lake Erie with the location of Station 4a, with the (e) zoomed view of the location of station. A representative obstructed satellite MODIS Image for this period is also shown (f).

can be seen that the edge of 0.8 ice concentration line is further pushed to the shore side (south east shore) in the fully coupled model compared to the 1-way coupled model. The shift in this edge line is consistent with the direction of the waves, which are shown by arrows in the magnified figure. In fact, the difference between the two edge lines are attributed to the ice breakage in the fully coupled model that are caused by the incoming waves. These waves break the ice edge as they encounter the ice floes and therefore, displace the ice concentration edge line further towards the south east shore side. The maximum difference between the two edge lines occurs at the center of the line and is about 3.5 km.

3.3. Daily lake-averaged ice cover

The lake-averaged ice cover, IC, represents the average concentration of ice covering a lake. It is calculated by normalizing the sum of ice concentrations over various control areas within the lake by the total area of the lake:

$$IC = \frac{\sum_{n=1}^{N} \left[(\text{Area})_n \times (\text{A}_{\text{ice}})_n\right]}{\text{Lake Area}}, \tag{21}$$

where $(\text{Area})_n$ is the area of the nth control volume of the lake, $(\text{A}_{\text{ice}})_n$ is the ice concentration in the nth control volume, N is the total number of control volumes considered, and Lake Area is the total area of the lake.

Fig. 12 illustrates the daily lake-averaged ice cover in different lakes from January to May 2011. Simulation results from different FVCOM models are compared against NIC observations. Results of the Model R4 is regenerated from the model presented by Bai et al. (2020). The fully coupled model predicts Lake Erie and Lake Huron with the highest ice cover averages during winter, at 60.6% and 36.9%, respectively, while Lake Ontario and Lake Superior have the lowest averages at 5.8% and 8.9%, respectively. The comparison shows significant improvement in the proposed model for ice cover prediction, particularly for Lake Superior and Lake Ontario. Table 2 displays the ice cover average values in each lake for the period of January–May 2011. These values are compared with the NIC observation data and the corresponding errors are also reported. Results show that the proposed model reduces the error of the ice cover for Lake Superior and Lake Ontario to 20% and 13.1%, respectively. Compared to the ice cover results predicted by the previous model of Bai et al. (2020) (137.5% for Lake Superior, and 49.6% for Lake Ontario), this corresponds to an improvement of 117.5% and 36.5% for Lake Superior and Lake Ontario, respectively. All the other lakes also show an improvement of more than 10% of error. This enhancement stems from pre-run calibration adjustments to melting coefficients, ice strength, wind forcing adjustments, and initializing the model with ice (see Appendix A for details). During this period, small improvements are also observed in the fully coupled model compared to the 1-way coupled model noting the ice breakage

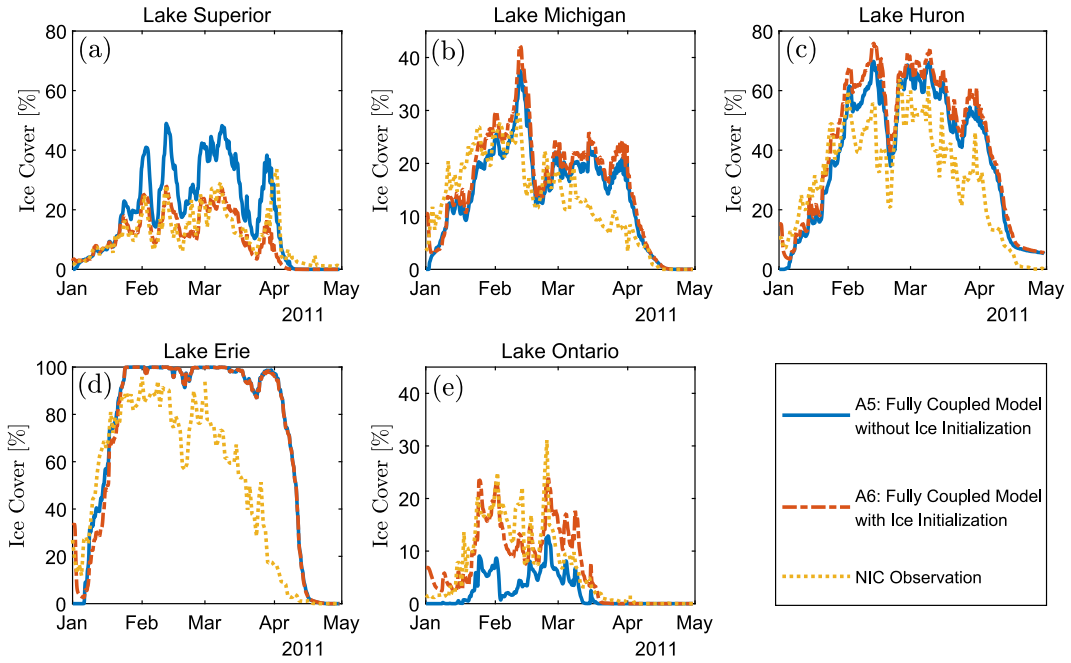


Fig. 14. Effect of ice Initialization on the daily lake-averaged ice cover. Results show that the ice initialization drastically improves the results for Lake Superior and Lake Ontario. Not only does the ice initializing improve the results in early days, but also it has lasting benefits for the entire winter.

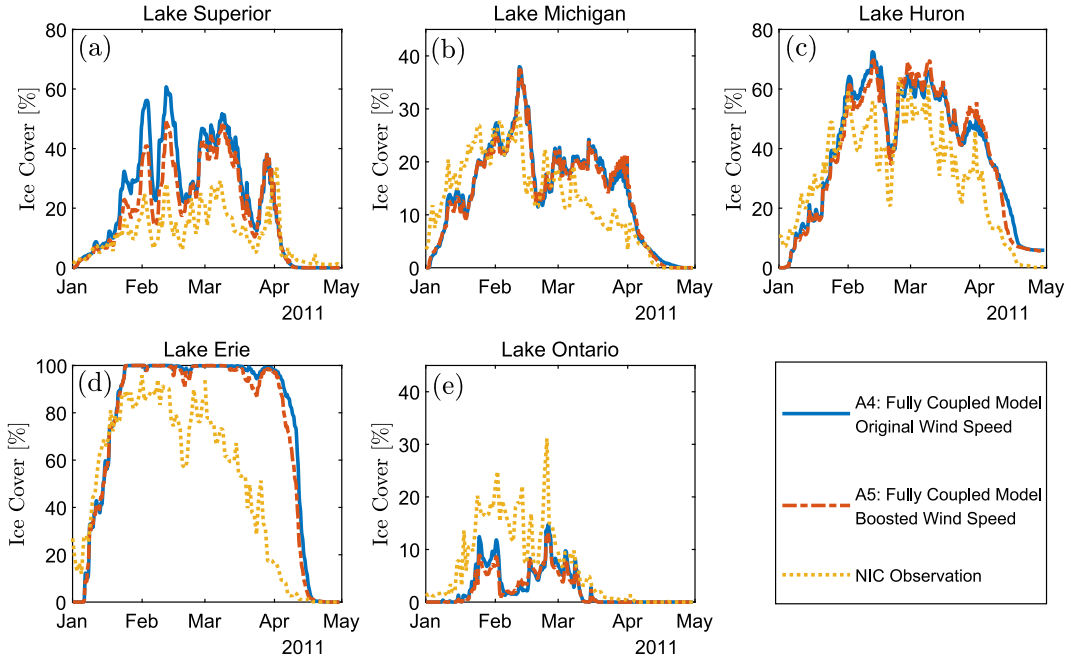


Fig. 15. Effect of boosted wind on the daily lake-averaged ice cover. Employing Boosted wind data as the input for the model improves the ice cover in Lake Superior, since this lake has larger wind fetch compared to the other lakes.

effects. As noted previously, the ice breakage effects are more prominent in the instantaneous (short term periods) or localized studies and are relatively minimal in the longer term and/or lake-averaged evaluations. While the improvements were measured with respect to the NIC observation data, it should be recalled that there is some uncertainty associated with the NIC products, as discussed in Section 3.1.

3.4. Application in Lake Erie

Previous studies by Hawley et al. (2018) provide rare winter measurements in the Great Lakes, specifically in Lake Erie. Fig. 13 compares predicted FVCOM results with NIC and Hawley et al.'s observations. Other than the fully and 1-way coupled models, results of the R4 model is also plotted that represents a model that is regenerated from (Bai et al., 2020) proposed ice-induced wave attenuation model. As shown in Fig. 13(a), the proposed FVCOM model notably improves ice concentration prediction at Station 4a compared to the original Bai et al. model. Both fully coupled and 1-way coupled models track the ice concentration reduction between February 18–22, albeit with a slightly delayed reduction and faster recovery compared to NIC observations. Several factors contribute to this deviation. Firstly, the FVCOM model in this study does not account for inflows, neglecting potential warming

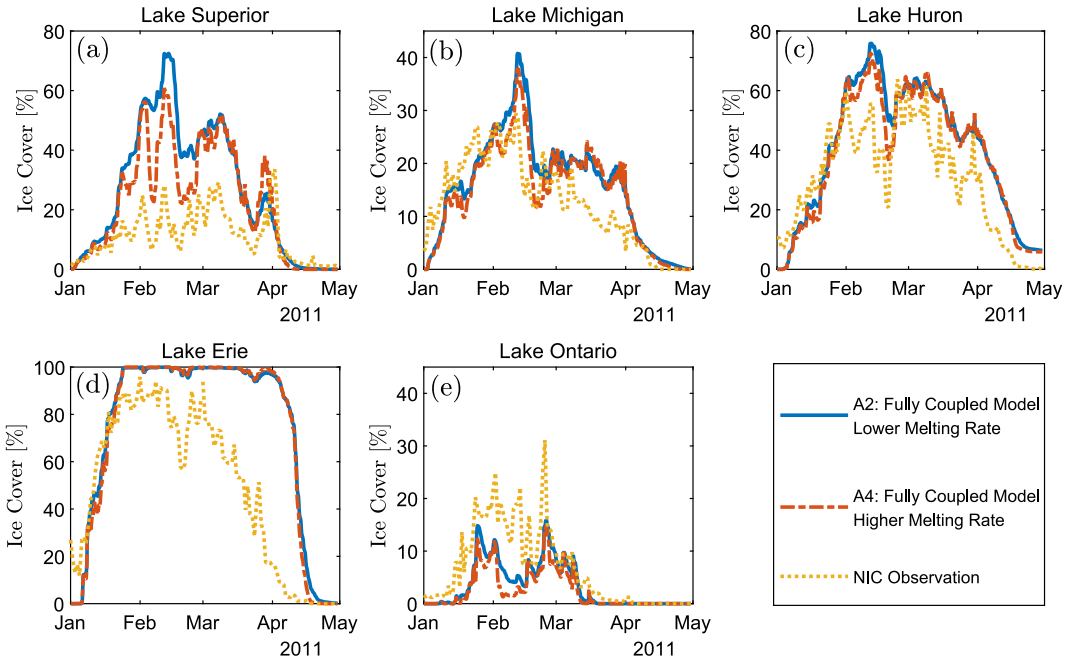


Fig. 16. Effect of melting rates on the daily lake-averaged ice cover. Increasing melting rate also results in improvements mainly in the Lake Superior.

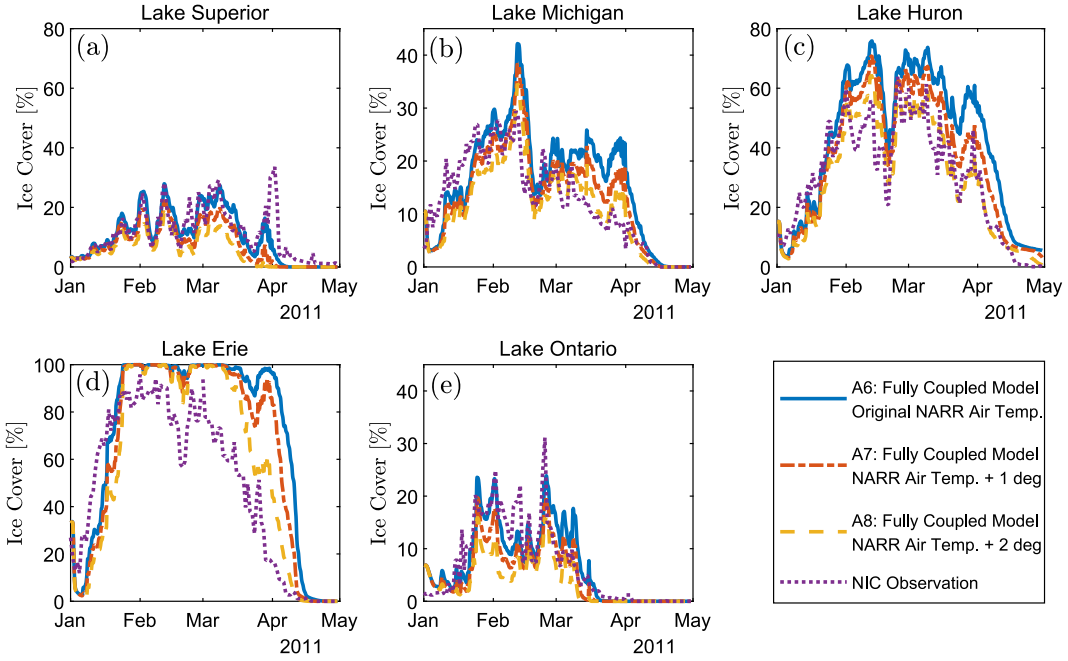


Fig. 17. Effect of air temperature on the daily lake-averaged ice cover. Increasing temperature data as the forcing input slightly reduces the ice cover in all the lakes. Cannon et al. (2023) showed that different lakes have different NARR temperature biases. Therefore, no modifications were considered for the runs of this study (R1, R2 and R3) and only original NARR air temperature data was utilized.

effects from southern river influxes. Additionally, NIC observations are derived from MODIS satellite images with low spatial resolution and significant interpolation that can introduce uncertainty. The spatial resolution can play a significant role in such cases. For example, location of Station 4a and neighboring nodes are shown in Figs. 13(d) and 13(e), respectively. The shades in Fig. 13 illustrate the variance of results in the neighboring nodes signifying the importance of the spatial resolution. Further analysis of MODIS images during February 19–22 reveals significant cloud cover (*e.g.*, Fig. 13(f)), which can complicate the data accuracy through temporal integration and interpolation. Air temperature variations during this period (Fig. 13(c)) align closely with fluctuations in predicted FVCOM ice concentration (Fig. 13(a)), reinforcing these uncertainties.

Bai et al. (2020) previously suggested that their model's deviation from NIC observations could stem from inadequate ice breakage capabilities. However, Fig. 13(a) indicates minimal ice concentration changes due to ice breakage (except a small reduction on February 21). Thus, calibration and ice initialization improvements, rather than ice breakage processes, primarily account for the predicted results' enhancement.

Fig. 13(b) compares FVCOM ice thickness results with daily averaged SWIP observations by Hawley et al. (2018). Again, results from

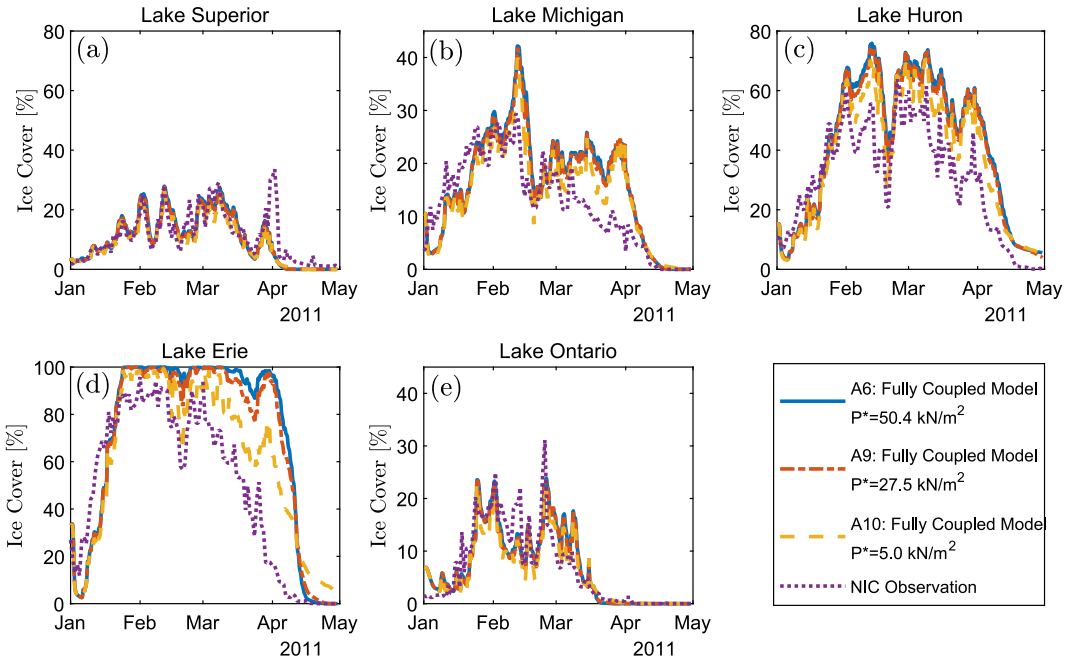


Fig. 18. Effect of ice strength on the daily lake-averaged ice cover. Reducing ice strength results in lower ice cover data in the lakes. The reduction is more prominent in Lakes Erie, Michigan and Huron. $P^* = 5$ kN/m^2 is a reasonable value for ice strength in the freshwater ice.

Table 2
Ice Cover for different lakes over the period of January–May 2011 resulted from different FVCOM models. Results are compared with the NIC observation data.

Model	Daily Ice Cover [–] and the Error [%] with respect to the NIC observation									
	Huron		Ontario		Michigan		Erie		Superior	
	Value	Error	Value	Error	Value	Error	Value	Error	Value	Error
NIC observation	30.2	–	6.7	–	12.5	–	49.0	–	11.2	–
Fully coupled	36.9	22.0	5.8	−13.1	14.1	12.7	60.6	23.6	8.9	−20.0
1-Way coupled	37.0	22.4	5.8	−13.1	14.2	13.3	60.6	23.8	9.0	−19.7
No coupling	37.2	22.9	6.0	−10.6	14.3	13.9	60.6	23.7	9.2	−17.9
Bai et al. Model	40.2	32.8	3.4	−49.6	15.7	25.1	73.7	50.5	26.5	137.5

Table 3
Numerical setup for the additional simulation runs. All the runs are conducted with the fully coupled model.

Run no.	Ice initialization	Wind speed	Forcing air temperature	Lateral melting Coef.[a]	Basal melting Coef.[b]	Ice strength[c]
A2	No	Original	NARR	1.6×10^{-6}	0.006	P* = 50.0
A4	No	Original	NARR	8.0×10^{-6}	0.030	P* = 50.0
A5	No	20% boosted	NARR	8.0×10^{-6}	0.030	P* = 50.0
A6	Yes	20% boosted	NARR	8.0×10^{-6}	0.030	P* = 50.0
A7	Yes	20% boosted	NARR+1 °C	8.0×10^{-6}	0.030	P* = 50.0
A8	Yes	20% boosted	NARR+2 °C	8.0×10^{-6}	0.030	P* = 50.0
A9	Yes	20% boosted	NARR	8.0×10^{-6}	0.030	P* = 27.5
A10	Yes	20% boosted	NARR	8.0×10^{-6}	0.030	P* = 5.00

[a] Unit in m/(s°C$^{1.36}$).
[b] Unitless.
[c] Unit in kN/m^2.

the 1-way coupled R4 model, derived from Bai et al. (2020), are also shown. The FVCOM models demonstrate reasonable ice thickness prediction, with the proposed model performing better than the original model. Both fully coupled and 1-way coupled models track the thickness reduction trend from February 13 to 22. It is speculated that the increase in the observed ice thickness from February 13 to 15 likely reflects measurement uncertainties rather than genuine physical changes. The trend of ice concentration (Fig. 13(a)) and air temperature results (Fig. 13(c)) support this speculation.

Conclusions

In this study, a fully coupled ice–wave interaction model was introduced that incorporates ice-induced wave attenuation and wave-induced ice breakage. The model was successfully implemented in the Finite-Volume Community Ocean Model (FVCOM) to enhance its utility in simulating complex oceanographic phenomena influenced by sea ice dynamics and wave energy. This was accomplished by integrating and coupling wave and ice modules to form the Great Lakes FVCOM–SWAVE–UG-CICE framework. The results of the model

were verified for the Great Lakes against observation data and field measurement conducted in the winter of 2011. The fully coupled model demonstrated robust predictive capabilities, particularly in simulating ice concentration and thickness, as well as significant wave heights, compared to observational data. Incorporating ice-induced wave attenuation and wave-induced ice breakage mechanisms significantly enhanced the understanding of ice dynamics in these large freshwater bodies. The findings also demonstrated that wave-induced ice breakage can have significant instantaneous and localized impacts. Specifically, the results revealed that incoming waves can break the ice and reduce the ice concentration by up to 10% in the marginal ice zones (MIZ). Furthermore, the analysis showed that wave-induced ice breakage can shift the MIZ edge, the boundary line where ice concentration reaches 80%, by up to 3.5 km in the direction of wave propagation.

The concept proposed in this study can be utilized for other models as well. For instance, the new stand-alone community version of CICE (version 6.0) has a joint floe-size/thickness distribution model, which can be coupled with a wave model for two-way interactions. Implementing this coupling and integrating it in the FVCOM would constitute a robust model, where the unstructured grid and finite-volume approach allow for precise representation and investigation of complex coastlines and bathymetries. Although the focus of this paper was on the Great Lakes, the setup offers similarity to many other enclosed basins subjected to freeze-up over the winter months (Javaherian et al., 2025). The proposed model in this study can be utilized with the ice–wave modeling and studies that focus on such regions (Iwasaki and Otsuka, 2021; Soomere, 2023; Yue et al., 2019).

The study also demonstrated the importance of model calibration, including adjustments to melting coefficients, ice initialization, and wind forcing inputs, which were crucial for improving the accuracy of ice predictions. For instance, the proposed model showed a consistent improvement in the predicted daily ice cover across the lakes compared to the previous models. This enhancement was specifically prominent for Lake Superior and Lake Ontario, where a reduction of error of 117.5% and 36.5% in the predicted daily ice cover was recorded, respectively.

Overall, this study proposes a robust fully coupled GL-FVCOM–SWAVE–UG-CICE model that provides valuable insights into the waves and ice interactions, crucial for environmental monitoring and management strategies in the ecologically sensitive regions. Future research directions should focus on refining model complexities, incorporating additional environmental variables, and expanding observational networks to further improve predictive accuracy and reliability in ice forecasting and climate impact assessments.

CRediT authorship contribution statement

M. Javad Javaherian: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **David Cannon:** Writing – review & editing, Supervision, Software, Project administration, Methodology, Funding acquisition, Conceptualization. **Jia Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Ayumi Fujisaki-Manome:** Writing – review & editing, Software, Resources, Methodology, Conceptualization. **Peng Bai:** Writing – review & editing, Software, Methodology, Conceptualization. **Lei Zuo:** Writing – review & editing, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Sensitivity analysis

Results of the sensitivity analysis are presented here to show the effects of the parameters that are calibrated and improved in this study. Table 3 reports the additional runs that are conducted with the fully coupled model to investigate the effects of the calibrated parameters. Figs. 14–18 illustrate these effects.

Data availability

Data will be made available on request.

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